

Chapter 8: Practical MLOps

Preparation For AI: From Raw Data To Reliable Models

Alexander D'hoore

Practical MLOps

The previous chapters prepared data.

Chapter 8 asks:

What happens after a model is trained?

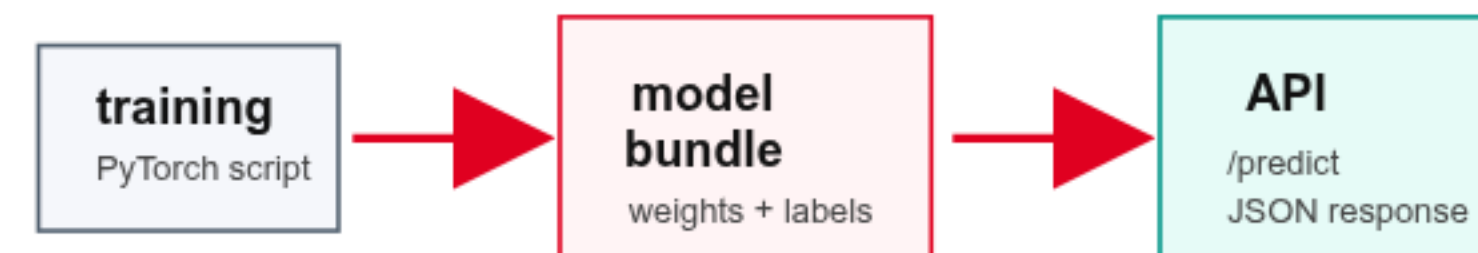
The answer is evidence plus a small API.



A Model Is Not Enough

A team needs to answer:

- what produced this model?
- which data trained it?
- how good was the run?
- where is the model bundle?
- can another system use it?



MLflow UI



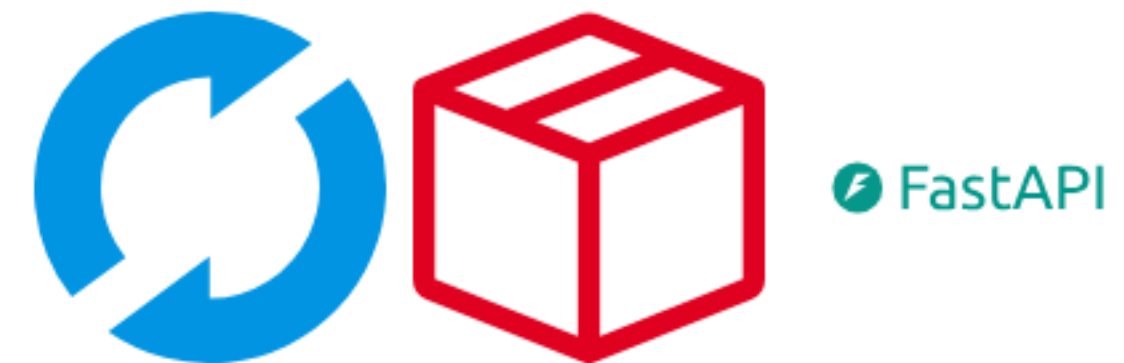
The Minimum Professional Handoff

This chapter stays narrow:

```
track the model training run  
save a usable model bundle  
serve predictions through a small API
```

MLflow handles the run evidence.

FastAPI handles the serving interface.

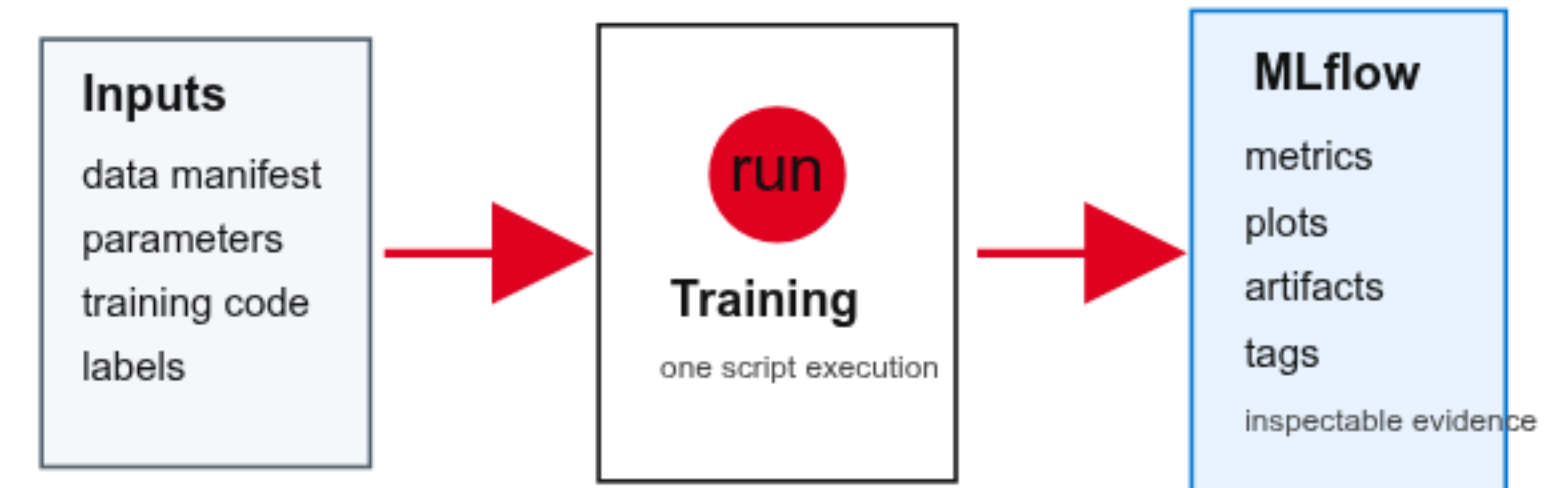


The Recurring Question

Teams keep asking:

Which run produced this model?

Good MLOps practice turns training into an inspectable engineering process.



A Useful Run Record

A useful run record contains:

- dataset location or version
- training parameters
- model family
- metrics over time
- evaluation artifacts
- model artifact



Accuracy Alone Is Thin Evidence

Two runs can both print:

```
validation accuracy: 0.82
```

But one run may use different labels, data, preprocessing, or trainable layers.

The metric needs context.



Parameters, Metrics, Artifacts

MLflow separates run evidence:

parameters: choices before training

metrics: measured behavior over time

artifacts: files that explain the result

That split makes runs easier to compare.

Dataset Evidence Still Matters

MLflow does not replace the data platform.

The run should point back to prepared data:

- manifest key
- selected labels
- train and validation counts
- object-storage location
- snapshot or dataset version

A model run should point back to the dataset definition that trained it.

Comparing Runs

MLflow becomes useful when there is more than one run.

In the lab:

run 1: frozen backbone

run 2: unfreeze one backbone block

Then compare before serving.

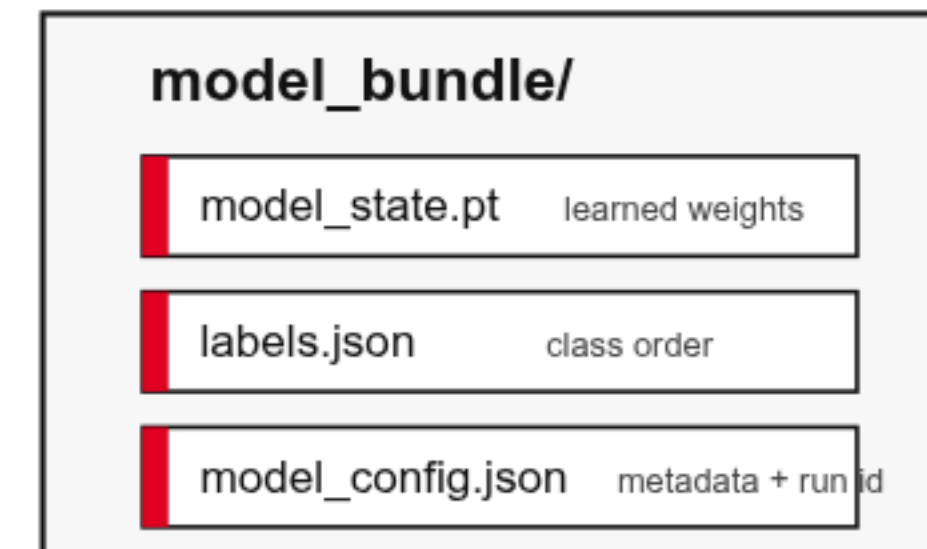


The Model Bundle

A trained model file alone is not enough.

The service also needs:

- architecture choice
- preprocessing rules
- label mapping
- expected input shape
- metadata
- MLflow run id



Weights without context are a weak handoff.

REST APIs

Training a model is not the same as using it.

A client sends an HTTP request.

The model service returns a JSON response.

```
client -> /predict -> model service -> JSON
```

The API is where the model becomes usable software.

API Responsibilities

FastAPI is the concrete API framework for this lab.

It automatically creates interactive docs.

The API should:

- validate the request
- apply the same preprocessing
- load a known bundle
- return predictable JSON
- expose health and metadata

Serving a model is software engineering, not only data science.

Predictable Response

The API returns structured JSON:

```
{  
  "predicted_label": "pizza",  
  "confidence": 0.91,  
  "model_name": "mobilenet_v3_small",  
  "mlflow_run_id": "..."  
}
```



Two Lives Of A Model

A production model has two visible lives:

- training evidence explains where the model came from
- service behavior explains what happens after deployment

Monitoring comes next:

- health
- latency
- failures
- input drift
- quality degradation



Lab Orientation

The labs reuse the Food-101 image classifier pattern.

New around the model:

```
log evidence to MLflow  
compare two runs  
choose a model bundle  
serve it with FastAPI
```



Lab Flow

The lab follows:

Food-101 manifest

- > tracked MLflow run
- > model bundle
- > FastAPI prediction service

Same data path, new evidence and handoff.



What To Notice In The Lab

Look for:

- selected labels
- logged parameters
- validation metrics
- artifacts
- model bundle contents
- API metadata
- prediction response

The point is an inspectable path from data to usable prediction.

Practical MLOps

Preparation for AI is not only training a model.

It is making the data and model path understandable enough that another person can:

- rerun it
- compare it
- serve it
- use it

That is the minimum professional handoff.